EXTENDED ABSTRACT

Design and Structural Analysis of a Precision XYZ Motion Platform for Ultrasonic Machine

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1. Introduction and Objectives

Single crystal diamond is widely used in precision optics, semiconductor devices, micromachining, and ultra-precision manufacturing due to its exceptional hardness, thermal conductivity, and wear resistance. However, these same properties make it difficult to machine using conventional methods, which often result in high cutting forces, excessive tool wear, and crack formation.

To overcome these limitations, non-conventional processes such as Ultrasonic Machining (USM) are used. USM removes material using high-frequency vibrations and abrasive slurry action, making it suitable for machining hard and brittle materials under low cutting forces.

The performance of ultrasonic machining depends not only on the machining process but also on the stability and accuracy of the supporting motion system. Therefore, the present work focuses on the design and simulation-based evaluation of a precision XYZ motion platform for ultrasonic machining applications using a CAD-based bottom-up design approach.

The main objectives of this work are:

- To design a precision XYZ motion platform for ultrasonic machining applications
- To develop and assemble individual machine components using CAD modelling
- To select suitable materials based on functional requirements of each component
- To perform static structural analysis under varying loading conditions
- To evaluate stress distribution, deformation, and Factor of Safety (FoS)
- To carry out modal analysis of the Z-axis and tool head assembly
- To study the structural and dynamic stability of the designed system

The overall aim of the project is to develop a structurally reliable and dynamically stable motion platform suitable for low-force ultrasonic machining applications involving brittle materials such as single crystal diamond.

2. Literature Review

Ultrasonic Machining (USM) is a non-conventional machining process widely used for hard and brittle materials such as ceramics, glass, carbides, and diamond. Unlike conventional machining, USM removes material through high-frequency vibration-assisted abrasive action, making it suitable for materials prone to cracking and brittle fracture.

In USM, the tool vibrates at frequencies of 20–40 kHz with small amplitudes. These vibrations are transmitted through an abrasive slurry, where abrasive particles remove material through micro-chipping and erosion. Previous studies show that parameters such as vibration frequency, amplitude, and abrasive properties significantly influence machining performance and surface finish.

Research has also highlighted the importance of horn and tool design in ultrasonic systems. The horn acts as a vibration amplifier and transfers ultrasonic energy from the transducer to the tool tip. Material selection is equally important due to cyclic loading and high-frequency vibrations. Titanium alloys are commonly preferred for horns because of their favourable fatigue and vibration characteristics, while structural and motion components require high stiffness, wear resistance, and dimensional stability.

With the advancement of computational tools, Finite Element Analysis (FEA) has become widely used for evaluating stress, deformation, and vibration behaviour before fabrication. However, most studies focus mainly on tool dynamics and machining behaviour, while limited work has been carried out on the structural design and analysis of complete ultrasonic machining platforms.

Therefore, the present work focuses on the design and structural evaluation of a precision XYZ motion platform for ultrasonic machining applications using CAD modelling and simulation-based analysis.

3. Methodology

The methodology adopted in this work focuses on the design, assembly, material selection, and simulation-based evaluation of a precision XYZ motion platform for ultrasonic machining applications. A bottom-up CAD modelling approach was followed using Fusion 360, where individual components were designed separately and later assembled into a complete motion platform.

The developed system consists of linear guide rails, support rods, threaded rods, bearings, couplers, spindle assemblies, structural frames, and the ultrasonic transducer-horn support region. Motion constraints and assembly relationships were applied between components to simulate realistic translational movement, while interference checks ensured smooth operation.

A precision threaded rod and linear guide mechanism were selected to achieve accurate positioning and controlled movement of the worktable. Since ultrasonic machining operates under low cutting forces, structural rigidity and positional accuracy were prioritized during the design stage.

Material selection was carried out based on functional requirements such as strength, stiffness, fatigue resistance, and wear resistance. AISI 52100 steel was selected for bearings due to its high hardness and fatigue life, while stainless steel 440C was used for support rods and threaded rods because of its strength and corrosion resistance. Mild steel was used for the frame to provide rigidity, aluminium alloy was selected for couplers to reduce rotational inertia, and titanium alloy was considered for the horn region because of its favourable vibration characteristics.

Static structural analysis was performed under loading conditions of 10 N, 20 N, 30 N, 100 N, and 1 kN to evaluate stress distribution, deformation, and Factor of Safety

(FoS). Modal analysis was also carried out on the Z-axis and tool head assembly to study the vibration behaviour and dynamic stability of the system under ultrasonic operating conditions.

Overall, the methodology integrates CAD modelling, material selection, assembly validation, and simulation-based analysis to develop a structurally stable and dynamically reliable ultrasonic machining platform.

4. Conclusion

The developed XYZ motion platform was successfully designed and analysed using a CAD and simulation-based engineering approach. Individual components such as guide rails, support rods, threaded rods, bearings, couplers, spindle assemblies, and structural members were modelled and assembled to form a complete precision motion system suitable for ultrasonic machining applications.

Static structural analysis performed under loading conditions ranging from 10 N to 1 kN showed that the structure experiences low stress and deformation under normal ultrasonic machining loads while maintaining a high Factor of Safety. The results confirmed that the platform possesses sufficient rigidity and structural stability for precision motion applications.

Modal analysis of the Z-axis and tool head assembly demonstrated stable dynamic behaviour within the ultrasonic operating frequency range. The obtained mode shapes and displacement patterns indicated that the structure is capable of supporting ultrasonic vibration-assisted machining operations without severe instability in the critical machining regions.

Overall, the study demonstrates the successful integration of CAD modelling, material selection, assembly validation, and simulation-driven analysis in the development of a precision XYZ motion platform for ultrasonic machining applications involving brittle materials such as single crystal diamond.